

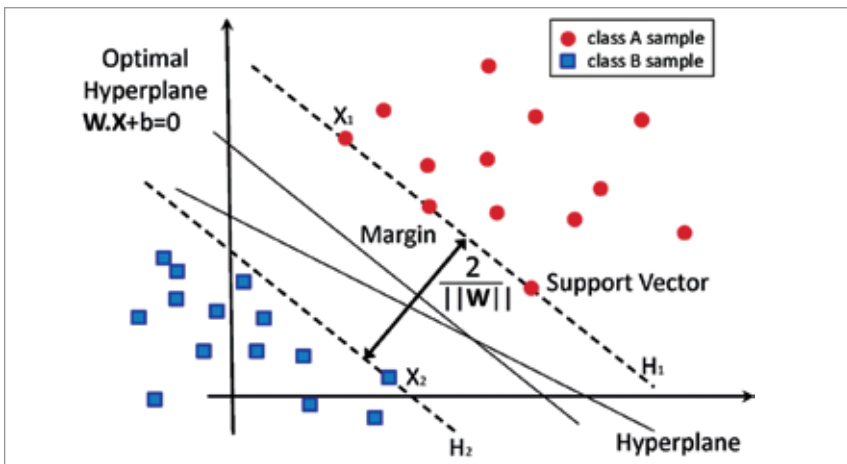


Fig. 4: Support Vector Machine algorithm (Courtesy: Google Images)

It is tempting to put an autonomous vehicle into the reinforcement learning box, but the reality is that given the complexity of the tasks that the self-driving car has to tackle, some form of each algorithm is used to tackle one or more of each type of problems.

Following are some commonly used ML algorithms by autonomous vehicles based on the environment or task:

Decision matrix algorithms.

These types of algorithms identify and analyse the relationships between different types of data in the given dataset. Based on these relationships, decision matrix algorithms come up with scores for each possible decision that could be made given the data. For instance, should the car press the brakes given the obstacle is 200 metres ahead? And if so, by what level of intensity should the brakes be pressed? Decision matrix algorithms help autonomous vehicles make such crucial decisions with a level of confidence that is calculated by the least probability of errors in the decision.

The Adaboost algorithm is a good example of a decision matrix implementation. As shown in Fig. 2, Adaptive Boosting or Adaboost is a pattern recognition algorithm that learns to differentiate between different patterns. It focusses on giving 'weight' or priority to its weakest links where ambiguity is the highest, until it can eliminate the maximum amount of errors possible in its decision making.

Clustering algorithms. As discussed previously, clustering is a form of unsupervised learning that helps

recognise difficult patterns from data points in datasets with high ambiguity. Even though self-driving vehicles make use of high-resolution, high density cameras, the environment, weather, lighting and many other such variables can affect the clarity and resolution of the images making object detection a difficult task. Clustering algorithms are specialised in recognising the objects in such a scenario. This can potentially save lives and make the vehicle super-efficient even in unfavourable situations.

A common clustering algorithm used for this purpose is the k-means clustering algorithm. K-means divides the data points into k number of clusters. It does so by using data points called centroids that are utilised to define each cluster. A data point is said to belong to a certain cluster given its affinity to the centroid of the respective cluster. With a number of iterations that include choosing centroids and assigning data points to them, the k-means clustering algorithm becomes efficient at allocating a given data point to a particular cluster. Thus, getting efficient at recognising particular objects in an image. Fig. 3 shows how k-means algorithms cluster data with each iteration.

Data reduction algorithms. The images that are obtained from different sensors in the autonomous vehicle contain a lot of noise. By noise we mean data that is irrelevant to the task at hand. For instance, a self-driving car may detect a cat crossing the street. But in order to detect that, a whole bunch of other data that the sensor

catches first needs to be disregarded and only the data related specifically to the cat on the road needs to be considered. In such scenarios, we use what are called data reduction algorithms that help in removing noise from the data and recognising relevant patterns.

Principal Component Analysis (PCA) and Support Vector Machine (SVM) are good examples of data reduction algorithms. For instance, in the case of SVM, we draw a hyper-plane between two distinct classes of data, thus differentiating between different patterns. With each iteration, the boundary is adjusted until the most optimal boundary can be achieved. This is done with the help of the data points at the boundaries that are referred to as support vectors as shown in Fig. 4.

Regression based algorithms.

As discussed previously, regression algorithms are a form of supervised learning algorithms that are trained on 'tagged' datasets with predefined independent and dependent variables. Based on the relationship between these variables, over a period of iterations, regression algorithms become efficient at predicting certain scenarios. In the case of the autonomous vehicles, the distance of a particular object can be measured in relation to the progressive change of size and the position of the object in each image incrementally.

To make this simpler, consider a traffic light. If the self-driving car takes an image of an upcoming traffic light from 500 metres away, the light will have a certain size and position in that image. But as the car gets closer to the light, say 200 metres away, the size and the position of the light in the latest images obtained by the car will be significantly larger. This form of data can be fed to a dataset on which regression can be applied to predict the distance of traffic light from the car. With this information, the car can decelerate accordingly.

Simple Linear Regression is a common example of regression analysis with which the value of a dependent variable can be predicted by defining its relationship with an independent variable. In a dataset if two variables show a strong positive/nega-